Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 10	**Date: 23/10/2026**	**Enrolment No:92400133079**

CODE

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Read the image in grayscale
img = cv2.imread("D:\SEM 3\python\(1).jpg", 0)

# Check whether the image was loaded successfully
if img is None:
    raise FileNotFoundError("Image not found. Check the image path.")

# 2. Apply the 2D Fourier Transform
F = np.fft.fftshift(np.fft.fft2(img))

# 3. Create the frequency distance matrix
M, N = img.shape

u, v = np.meshgrid(
    np.arange(N) - N // 2,
    np.arange(M) - M // 2
)

D = np.sqrt(u**2 + v**2)

# 4. Create Low-Pass and High-Pass Filters
LPF = D < 30
HPF = D >= 30

# 5. Apply the filters and compute the inverse FFT
smooth = np.abs(
    np.fft.ifft2(np.fft.ifftshift(F * LPF))
)

high_pass = np.abs(
    np.fft.ifft2(np.fft.ifftshift(F * HPF))
)


# Sharpen the image by adding high-frequency components
```

Marwadi University Faculty of Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 10	**Date: 23/10/2026**	**Enrolment No:92400133079**

```
sharp = np.clip(img.astype(float) + high_pass, 0, 255)

# 6. Display the original, smoothed, and sharpened images
plt.figure(figsize=(12, 4))

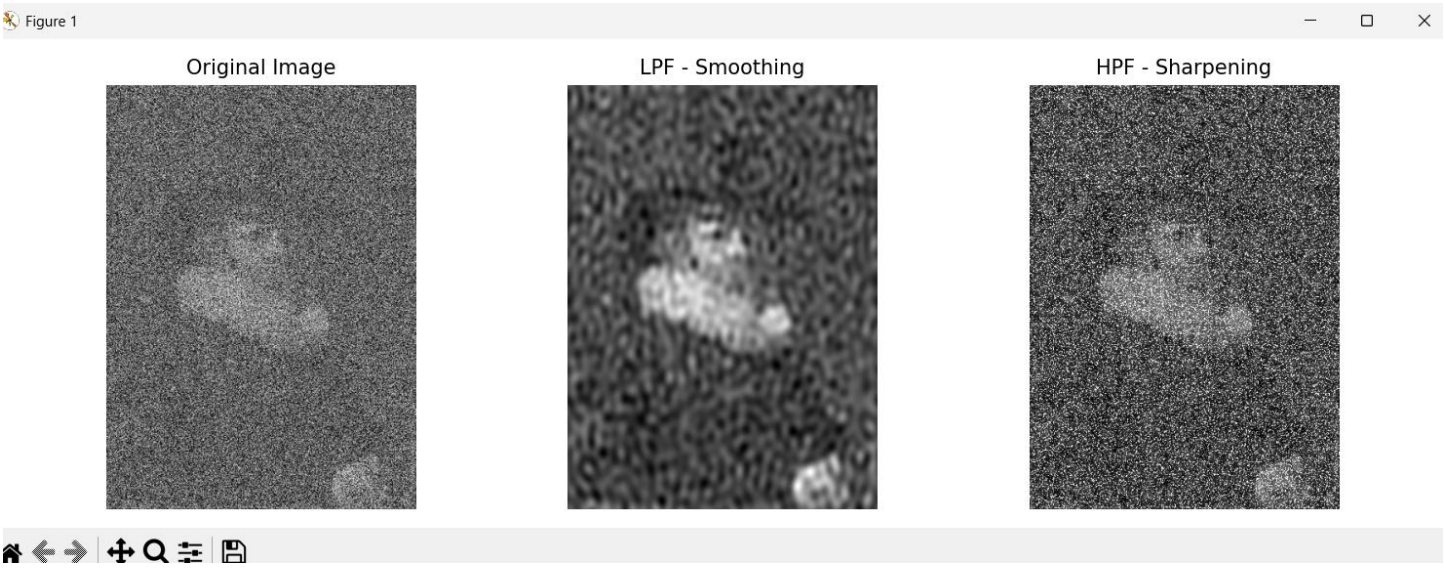
plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothing')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')

plt.tight_layout()
plt.show()
```

OUTPUT



GITHUB : https://github.com/yugmakwana99/DSIP.git

Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 10	Date: 23/10/2026	Enrolment No:92400133079